Orbital Collapse Architecture: A Power-Law Model of Institutional Absorption Following Systemic Financial Failure

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Elán Zainos Corona · Fractal Core Research, Tlaxcala, Mexico · ORCID 0009-0009-9125-253X

Abstract

When a dominant financial institution fails, its assets, counterparties, and market position do not dissolve — they are *absorbed* by an identifiable successor, and the speed of that absorption is empirically regular. We model the post-failure dominance ratio $R(t) = mass_{absorber}(t) / mass_{collapsed,peak}$ as a power law, $R(t) = a \cdot t^b$, where the exponent **b** measures the velocity of institutional absorption following functional extinction of the failed hub.

Applying this Orbital Collapse Architecture (ACO) framework to the six canonical absorptions of the 2008–2009 crisis — Lehman Brothers → Barclays + JPMorgan, Bear Stearns → JPMorgan, Washington Mutual → JPMorgan (FDIC), Wachovia → Wells Fargo, Merrill Lynch → Bank of America, and Chrysler → Fiat + US Treasury — we recover statistically significant power-law absorption curves in every case (R^2 = 0.85–0.99, all $p < 0.01$), reconstructed entirely from primary sources (SEC EDGAR, Federal Reserve Flow of Funds, FDIC, SIGTARP, and the Valukas Examiner Report).

The central regulatory finding is that **institutional friction governs absorption velocity**: abrupt, disorderly failures (Lehman) and regulator-brokered resolutions (WaMu, FDIC) produce distinct absorption signatures, and the framework distinguishes genuine absorption from obsolescence-without-capture. These signatures offer a quantitative, falsifiable lens on too-big-to-fail concentration dynamics and on how resolution policy shapes the speed and concentration of post-crisis market restructuring — directly relevant to financial-stability risk and the design of orderly-resolution regimes.

The financial corpus is a domain-specific extract of a larger 721-case, cross-domain validation of the underlying satellization model, ensuring the mechanism is not financial-sector-specific but a general feature of coupled dominance dynamics.

Six financial cases (primary-source, v2.4.0)

Hub → Absorber	Trigger	b	R^2	p
Lehman Brothers → Barclays + JPMorgan	abrupt (2008)	+0.246	0.892	<0.001
Bear Stearns → JPMorgan	abrupt (2008)	+0.043	0.926	0.002
Washington Mutual → JPMorgan (FDIC)	abrupt (2008)	+0.009	0.946	0.001
Wachovia → Wells Fargo	abrupt (2008)	+0.153	0.892	0.004
Merrill Lynch → Bank of America	abrupt (2008)	+0.217	0.846	0.005
Chrysler → Fiat + US Treasury (TARP)	abrupt (2009)	+0.138	0.990	<0.001

All six cases reproduced from primary sources via `reconstruction_real/code/build_aco_v29.py`. Full corpus and code: github.com/Inzainos/The-shadow-Node-Theory

Relevance to the conference theme

- **Financial-stability risk:** quantifies post-failure concentration (the too-big-to-fail consolidation that followed 2008).
- **Market structure / resolution policy:** absorption velocity (b) is shaped by whether a failure is disorderly or regulator-brokered — a direct policy lever.
- **Empirical and reproducible:** every case is traceable to SEC / FDIC / Fed / SIGTARP primary records; nothing synthetic.